

```
In [23]: # Fill in student ID and name
#
student_id = "s223532749"
student_first_last_name = "Gordon Tse"
print(student_id, student_first_last_name)
```

s223532749 Gordon Tse

```
In [ ]: # Install libraris, if not yet.
# ! pip install firebase_admin pandas

import firebase_admin

databaseURL = 'https://sit225-d51dc-default-rtdb.asia-
southeast1.firebaseio.com/'
cred_obj = firebase_admin.credentials.Certificate(
    'sit225-d51dc-firebase-adminsdk-fbsvc-bb6c4788ce.json'
)
default_app = firebase_admin.initialize_app(cred_obj, {
    'databaseURL':databaseURL
})
```

```
-----
-----
ValueError                                Traceback (most
recent call last)
Cell In[24], line 10
      6 databaseURL = 'https://sit225-d51dc-default-rtdb.asia-
southeast1.firebaseio.com/'
      7 cred_obj = firebase_admin.credentials.Certificate(
      8     'sit225-d51dc-firebase-adminsdk-fbsvc-
bb6c4788ce.json'
      9 )
--> 10 default_app = firebase_admin.initialize_app(cred_obj, {
      11     'databaseURL':databaseURL
      12     })

File c:\Users\gordo\anaconda3\Lib\site-
packages\firebase_admin\__init__.py:74, in
initialize_app(credential, options, name)
      71     return app
      73 if name == _DEFAULT_APP_NAME:
--> 74     raise ValueError((
      75         'The default Firebase app already exists. This
means you called '
      76         'initialize_app() more than once without
providing an app name as '
      77         'the second argument. In most cases you only
need to call '
      78         'initialize_app() once. But if you do want to
initialize multiple '
      79         'apps, pass a second argument to
initialize_app() to give each app '
      80         'a unique name.'))
      82 raise ValueError(
      83     f'Firebase app named "{name}" already exists. This
means you called '
      84     'initialize_app() more than once with the same app
name as the '
      85     'second argument. Make sure you provide a unique
name every time '

```

86 'you call initialize_app().')

ValueError: The default Firebase app already exists. This means you called initialize_app() more than once without providing an app name as the second argument. In most cases you only need to call initialize_app() once. But if you do want to initialize multiple apps, pass a second argument to initialize_app() to give each app a unique name.

```
In [25]: from firebase_admin import db

# A reference point is always needed to be set
# before any operation is carried out on a database.
#
ref = db.reference("/")

# JSON format data (key/value pair)
data = { # Outer {} contains inner data structure
    "Book1":
        {
            "Title": "The Fellowship of the Ring",
            "Author": "J.R.R. Tolkien",
            "Genre": "Epic fantasy",
            "Price": 100
        },
    "Book2":
        {
            "Title": "The Two Towers",
            "Author": "J.R.R. Tolkien",
            "Genre": "Epic fantasy",
            "Price": 100
        },
    "Book3":
        {
            "Title": "The Return of the King",
            "Author": "J.R.R. Tolkien",
            "Genre": "Epic fantasy",
            "Price": 100
        },
    "Book4":
        {
            "Title": "Brida",
            "Author": "Paulo Coelho",
            "Genre": "Fiction",
            "Price": 100
        }
}

# JSON format data is set (overwritten) to the reference
# point set at /, which is the root node.
#
ref.set(data)
```

In [26]: `ref = db.reference("/") # set ref point`

```
# query all data under the ref
books = ref.get()
print(books)
print(type(books))

# print each item separately
for key, value in books.items():
    print(f"{key}: {value}")

# Query /Book1
ref = db.reference("/Book1")
books = ref.get()
print(books)
```

```
{'Book1': {'Author': 'J.R.R. Tolkien', 'Genre': 'Epic fantasy',
'Price': 100, 'Title': 'The Fellowship of the Ring'}, 'Book2':
{'Author': 'J.R.R. Tolkien', 'Genre': 'Epic fantasy', 'Price':
100, 'Title': 'The Two Towers'}, 'Book3': {'Author': 'J.R.R.
Tolkien', 'Genre': 'Epic fantasy', 'Price': 100, 'Title': 'The
Return of the King'}, 'Book4': {'Author': 'Paulo Coelho', 'Genre':
'Fiction', 'Price': 100, 'Title': 'Brida'}}
<class 'dict'>
Book1: {'Author': 'J.R.R. Tolkien', 'Genre': 'Epic fantasy',
'Price': 100, 'Title': 'The Fellowship of the Ring'}
Book2: {'Author': 'J.R.R. Tolkien', 'Genre': 'Epic fantasy',
'Price': 100, 'Title': 'The Two Towers'}
Book3: {'Author': 'J.R.R. Tolkien', 'Genre': 'Epic fantasy',
'Price': 100, 'Title': 'The Return of the King'}
Book4: {'Author': 'Paulo Coelho', 'Genre': 'Fiction', 'Price':
100, 'Title': 'Brida'}
{'Author': 'J.R.R. Tolkien', 'Genre': 'Epic fantasy', 'Price':
100, 'Title': 'The Fellowship of the Ring'}
```

In [27]: `# Write using push() function`

```
# Note that a set() is called on top of push()
#
ref = db.reference("/")
ref.set({
    "Books":
        {
            "Best_Sellers": -1
        }
})

ref = db.reference("/Books/Best_Sellers")

for key, value in data.items():
    ref.push().set(value)
```

```
In [28]: # Update data
#
# Requirement: The price of the books by
# J. R. R. Tolkien is reduced to 80 units to
# offer a discount.
#
ref = db.reference("/Books/Best_Sellers/")
best_sellers = ref.get()
print(best_sellers)
for key, value in best_sellers.items():
    if(value["Author"] == "J.R.R. Tolkien"):
        value["Price"] = 90
        ref.child(key).update({"Price":80})

{'-Oyquin0tTFaJI1rm1GK': {'Author': 'J.R.R. Tolkien', 'Genre':
'Epic fantasy', 'Price': 100, 'Title': 'The Fellowship of the
Ring'}, '-OyquirFUS24HgPGUAjn': {'Author': 'J.R.R. Tolkien',
'Genre': 'Epic fantasy', 'Price': 100, 'Title': 'The Two Towers'},
'-OyquivKF_9LBVnsEz7r': {'Author': 'J.R.R. Tolkien', 'Genre':
'Epic fantasy', 'Price': 100, 'Title': 'The Return of the King'},
'-OyquizJown7vK-kl0zy': {'Author': 'Paulo Coelho', 'Genre':
'Fiction', 'Price': 100, 'Title': 'Brida'}}
```

```
In [29]: # Let's delete all best seller books
# with J.R.R. Tolkien as the author.
#
ref = db.reference("/Books/Best_Sellers")

for key, value in best_sellers.items():
    if(value["Author"] == "J.R.R. Tolkien"):
        ref.child(key).set({})
```

```
In [30]: # Delete all best_seller data.
#
ref = db.reference("/Books/Best_Sellers/")
best_sellers = ref.get()
print(best_sellers)
print(type(best_sellers))

{'-OyquizJown7vK-kl0zy': {'Author': 'Paulo Coelho', 'Genre':
'Fiction', 'Price': 100, 'Title': 'Brida'}}
<class 'dict'>
```

```
In [31]: ref = db.reference("/Books/Best_Sellers")
ref.set({})
```